



ISBM College of Engineering, Nande, Pune

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Sales and Marketing Agent Development for Online Websites

An AI-Powered Multi-Agent System for Sales, Marketing and Lead Qualification

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Project Concept & Overview

Core Concept

Concept Statement

An intelligent B2B web system that replaces static forms and generic single-prompt chatbots with specialized, autonomous AI agents powered by LLMs and Retrieval-Augmented Generation (RAG).



Sales Agent

Handles customer inquiries, provides context-grounded consultation, presents pricing packages, and assists potential leads directly.



Marketing Agent

Generates targeted promotional content, creates personalized follow-up campaigns, and crafts customized business outreach emails.



Lead Qualification Agent

Analyzes incoming prospect details, evaluates budget and timeline, scores conversion readiness, and updates CRM tables.

Problem Statement & Motivation

Review Q1, Q2, Q3 & Q7

Current Industry Challenges

Manual handling of technical business inquiries, lead scoring, and campaign copy leads to delayed response times, high operational overhead, and lost sales opportunities.

The Rule-Based Chatbot Gap

Traditional rule-based website chatbots provide rigid, generic responses and struggle with multi-turn compliance standards (e.g., ISO 9001, IATF 16949 QMS consulting).

Concise Problem Statement

"To develop an autonomous AI multi-agent web platform that provides domain-grounded pre-sales consultations using RAG, automatically qualifies high-intent leads, and generates personalized marketing campaigns with zero manual delay."

- ✓ **Reduce Manual Pre-Sales Effort:** Automates routine Q&A and qualification.
- ✓ **Improve Response Speed:** Instant 24/7 technical answers keep leads engaged.

Similar type of methodology

LLM-Based Multi-Agents

Specialized agents collaborate through planning, communication and task execution.

Multi-Agent Chatbots

RAG, specialized agents and intent routing support customer interaction.

MaRGen

Multi-agent collaboration supports automated market research and analysis.

AI Salesman

LLM-driven sales dialogue and strategic planning improve sales interaction.

Project direction: combine specialized agents + LLMs + RAG for practical sales, marketing and lead automation.

Paper / System Title	Technique Used	Key Takeaway & Relevance
LLM-based Multi-Agents Survey IJCAI 2024	LLM Planning & Agent Communication	Provides foundation for orchestrating specialized AI agents with discrete functional roles.
Multi-Agent Chatbot Framework ScienceDirect 2026	RAG & Intent Routing	Demonstrates how RAG prevents LLM hallucinations in complex compliance customer support.
MaRGen: Market Research Agents arXiv 2025	Multi-Agent Collaboration	Validates automated market data retrieval and personalized campaign content drafting.
AI Salesman: Telemarketing LLM AAAI 2026	Dialogue Strategy Optimization	Explores structured sales dialogue strategies that guide users towards booking consultations.
LEAD-MAS: Business Lead Intelligence SSRN 2026	LLM-based Multi-Agent Lead Intelligence & Lead Scoring	Automates business lead discovery, information extraction, lead qualification, scoring, and CRM integration for prioritizing potential customers.

Conclusion: Extensive research literature validates our architecture (Multi-Agent + LLM + RAG + FastAPI).

Project Objectives

Review Q6, Q8 & Q9

🎯 Objective 1: Multi-Agent Automation

Automate 24/7 web pre-sales consultation and customer interaction using specialized LLM agents.

🎯 Objective 2: Lead Qualification

Automatically score incoming prospects based on extracted budget, timeline, and compliance needs.

🎯 Objective 3: Marketing Content Generation

Generate tailored promotional copy, follow-up emails, and client pitch decks automatically.

🎯 Objective 4: RAG Grounding

Integrate vector databases to provide accurate, hallucination-free ISO/QMS standards advice.

🎯 Objective 5: Centralized CRM Management

Store and sync lead profiles, interaction logs, and analytics into a secure owner portal.

🎯 Objective 6: Operational Efficiency

Reduce manual human pre-sales workload by over 70% and lower average response times to < 3 seconds.

System Methodology & Workflow

Methodology

1

Website Query

Visitor submits inquiry via online web chatbot interface.

2

Intent Routing

Classifier routes query to Sales, Marketing, or Lead Agent.

3

RAG Retrieval

Embeddings match query against ChromaDB knowledge base.

4

Agent Processing

LLM formulates context-aware response or marketing copy.

5

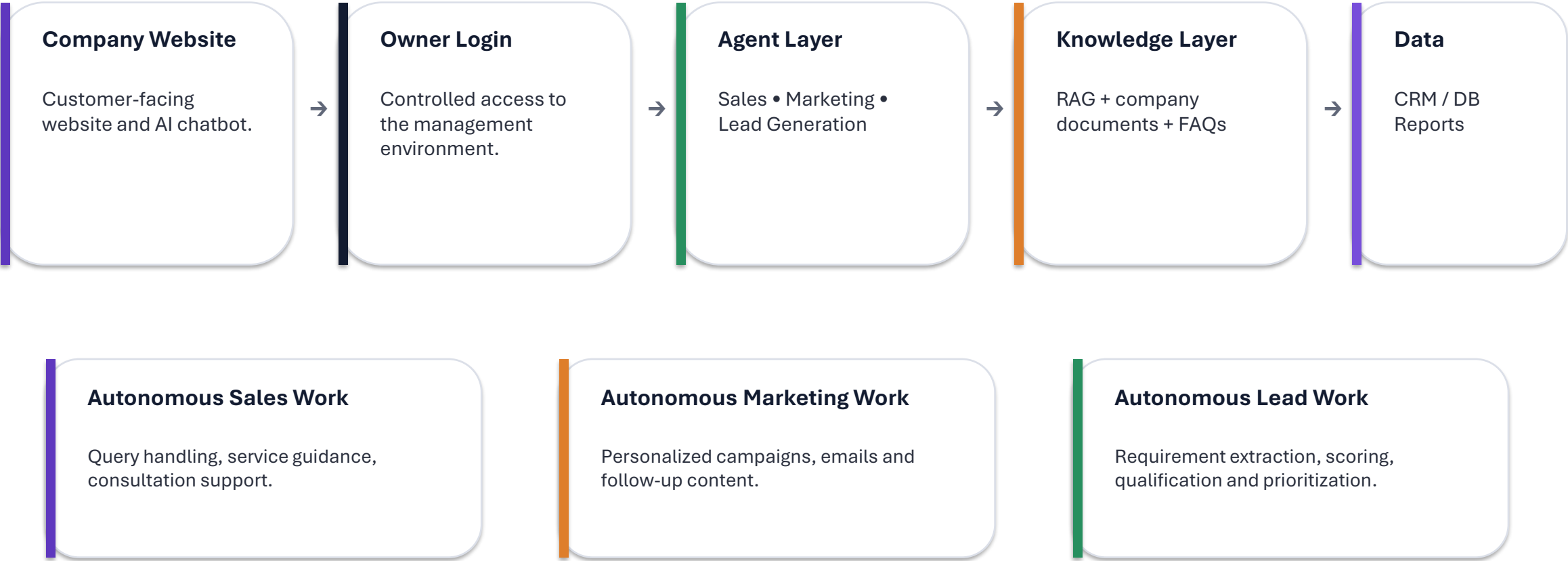
Output & CRM

Delivers answer, updates database, and alerts owner portal.

Methodological Validation

The system decouples user interaction from heavy backend computation. Intent routing ensures queries are processed by the most qualified specialized agent, while RAG restricts answers to verified company documentation.

Proposed System Architecture



Autonomous Agent Responsibilities

Agent Roles



Sales Agent

- Answers technical product & compliance standard FAQs.
- Explains service packages & consultancy pricing.
- Guides prospects to book appointment slots.



Marketing Agent

- Creates personalized cold outreach email copy.
- Generates promotional pitch decks & campaign content.
- Analyzes target audience industry profiles.



Lead Gen Agent

- Collects user contact details & compliance needs.
- Calculates lead quality & conversion scores.
- Updates PostgreSQL / CRM database tables.

System Inputs & Outputs Matrix

➡ System Inputs

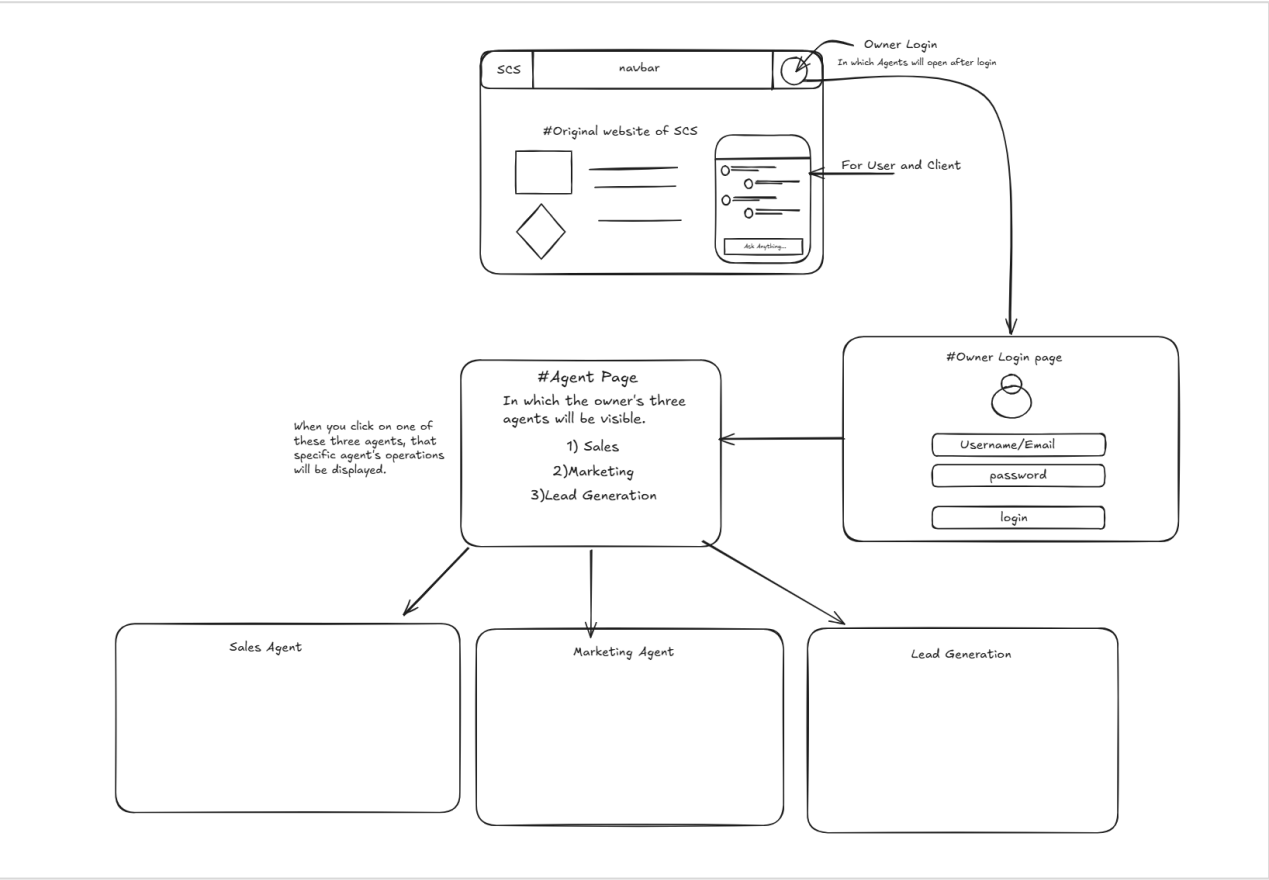
- Natural Language Queries:** Client questions regarding QMS, ISO certification, and timelines.
- Lead Profile Data:** Contact name, business email, company size, and budget parameters.
- Knowledge Documents:** Company PDFs, service guidelines, ISO compliance standards, and FAQs.
- Marketing Objectives:** Target industry verticals, promotional campaign goals, and tone preferences.

➡ System Outputs

- Context-Aware AI Answers:** Grounded technical answers delivered directly in chat.
- Lead Qualification Summaries:** Prioritized score rankings and conversion probability reports.
- Marketing Campaign Assets:** Customized follow-up emails, pitch messaging, and proposals.
- CRM & Analytics Entries:** Updated lead tables and scheduled calendar meetings for owners.

System UI Implementation & Prototype

Working UI



1. Client Web Chat Widget

Clean modern landing page for Samrath Consultancy featuring an integrated interactive AI Assistant.

2. Owner Portal Login

Secure authentication dashboard allowing business owners to monitor leads and agent operations.

3. Agent Workspace

Real-time pipeline metrics (58 New Leads, 11 Converted) and instant research tools for sales managers.

Technology Stack & System Specs

Software Architecture

Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	FastAPI (Python 3.11+)
AI Framework	LangChain, LangGraph / CrewAI
LLM & RAG	OpenAI GPT-4o / Gemini, ChromaDB
Database	PostgreSQL / MongoDB
Developer Tools	VS Code, Postman, Git, GitHub

Operating Specifications

Hardware	Intel i5 (11th Gen+) / 8GB+ RAM / SSD
Security	JWT Token Auth, Password Hashing
APIs	RESTful endpoints, Google Calendar API
Deployment	Uvicorn / Docker Containerized
Browsers	Chrome, Edge, Firefox, Safari

Project Review Checklist Alignment

Compliance

Q1: Research Gap

Identified limitations of static forms and generic rule-based chatbots.

Q2–Q3: Problem Statement

Clear, concise definition of multi-agent pre-sales & marketing automation.

Q4: Literature Survey

4 recent high-impact publications surveyed (IJCAI, ScienceDirect, AAAI).

Q5: Scope & Survey

Survey confirms multi-agent RAG architecture is optimal for domain Q&A.

Q6: Objectives Alignment

6 unambiguous goals mapped directly to project system deliverables.

Q7: Scope Clarity

Scope bounded to web chatbot, lead qualification, and owner portal.

Q8–Q9: Goal Understanding

System architecture and agent routing fully planned and documented.

Q10: Social Impact

Democratizes ISO compliance and reduces SME manual pre-sales cost.

Status Summary

100% Review-I Criteria Fully Addressed & Cleared.

Conclusion

The proposed **Sales and Marketing Agent Development for Online Websites** aims to make sales and marketing activities easier by using separate AI agents for different tasks. The **Sales Agent, Marketing Agent, and Lead Generation Agent** can handle their respective work and reduce repetitive manual effort.

The system also uses **RAG and company-specific knowledge** to provide more relevant responses. With the website, chatbot, owner login, and agent dashboard, the project provides a single platform for managing customer interaction, marketing activities, and leads.

Overall, the project can help make the **sales and marketing process faster, more organized, and easier to manage**, while also providing scope for future improvements such as CRM integration, analytics, and additional AI agents.

- 1) **O. A. Arslan and M. Mete**, *“Application Level Multi-Agent Methodology for Chatbots,”* Procedia Computer Science, vol. 280, pp. 807–815, 2026.
- 2) **T. Guo et al.**, *“Large Language Model based Multi-Agents: A Survey of Progress and Challenges,”* Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), 2024.
- 3) **R. Koshkin et al.**, *“MaRGen: Multi-Agent LLM Approach for Self-Directed Market Research and Analysis,”* LLM4ECommerce Workshop at KDD '25, 2025.
- 4) **A. Panda, B. Das, and N. Singh**, *“AI-Powered Sales and Marketing Analytics Platform Using Multi-Agent AI,”* International Journal of Data Science and IoT Management System (IJDIM), 2026.

Thank You!

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